

Practice Quiz 03D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 03

Constructing a counterexample to a quantified equivalence.

1. Let the universe of discourse be $D = \{0, 1\}$. Define predicates $P(x)$ and $Q(x)$ on D so that

$$\forall x \in D (P(x) \vee Q(x))$$

is true, but

$$(\forall x \in D P(x)) \vee (\forall x \in D Q(x))$$

is false. Give a two-row table listing $P(x)$ and $Q(x)$ for $x = 0, 1$, and use it to verify both claims.

Solution Let $P(x)$ mean $x = 0$, and let $Q(x)$ mean $x = 1$.

x	$P(x)$	$Q(x)$	$P(x) \vee Q(x)$
0	T	F	T
1	F	T	T

The disjunction $P(x) \vee Q(x)$ holds for both elements of D , so its universal quantification is true. But $\forall x \in D P(x)$ is false at $x = 1$, and $\forall x \in D Q(x)$ is false at $x = 0$. Therefore the second displayed statement is false. ■